

Comparing the Binding Affinity of a β 2AR Variant Using Density-Threshold Affinity (DTA) for Future Application to Different T-cell Receptor Conformation

Abstract:

Changes in protein structure can alter cholesterol (CHOL) binding sites, affecting protein-lipid interactions [1,2]. The β 2-adrenergic receptor (β 2AR) contains proposed CHOL-binding sites that may contribute to receptor stability [2]. We use a β 2AR variant missing α -helix ($\Delta\alpha 6$ variant) to test whether Density-Threshold Affinity (DTA) [3] can detect changes in CHOL-binding affinity, despite using the same accessible site (A_{site}). Compared with published results, site 1 showed similar affinity, while site 2 showed substantially reduced affinity. These results benchmark DTA's ability to quantify lipid enrichment and will be useful when applying the method to different T-cell receptors' (TCR) conformations.

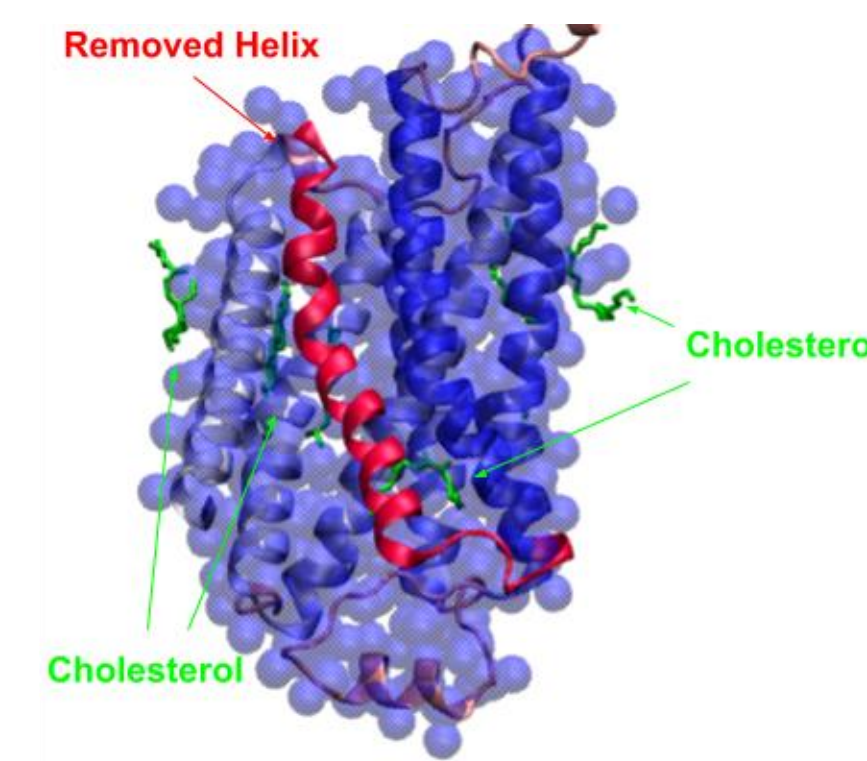
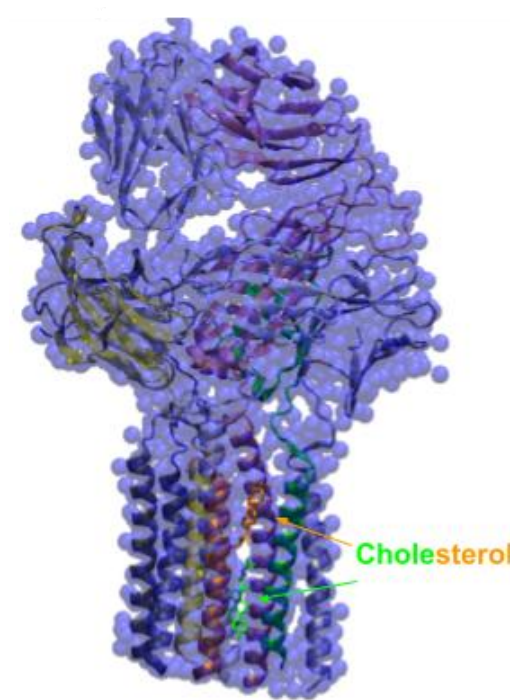


Fig.1: β 2AR variant structure and CG model, highlighting CHOL and the missing transmembrane helix.

Research Questions:

1. Can DTA distinguish differences in lipid-binding affinity between β 2AR variants despite using the same A_{site} ?
2. Do differences in protein structure correspond to differences in calculated binding affinity for CHOL?
3. Can the DTA workflow be applied to different TCR conformations to identify changes in lipid-binding affinity due to differences in protein structure for future application?

Fig. 2: TCR structure and CG model, highlighting CHOL molecules.



Background:

- **Membrane lipids influence protein structure, stability, and function** [1].
- **β 2AR** has two proposed CHOL-binding sites, making it a useful **DTA validation system** [2].
- **DTA** uses coarse-grain (CG) simulations to identify regions of lipid enrichment relative to the bulk membrane and quantify lipid occupancy at these sites. This can then be used to calculate **lipid-binding affinity (ΔG_{bind})** [3].
- **TCR** has two proposed CHOL-binding sites and can localize to **CHOL-enriched environments** associated with inhibition [4,5].
- **TCR conformation may alter CHOL-binding site location and affinity** [4].

Results

Table 1: DTA results for $\Delta\alpha 6$ β 2AR vs. published β 2AR [3]. Site 1 showed similar CHOL-binding affinity, while site 2 showed substantially reduced affinity.

Binding Affinity	Site 1	Site 2
Published β 2AR ΔG_{bind} (kcal/mol) [3]	-0.7 ± 0.4	-1.4 ± 0.2
$\Delta\alpha 6$ β 2AR ΔG_{bind} (kcal/mol)	-0.77	-0.16
Percent Error	10%	89%

Depletion 0.1 1 10 Enrichment

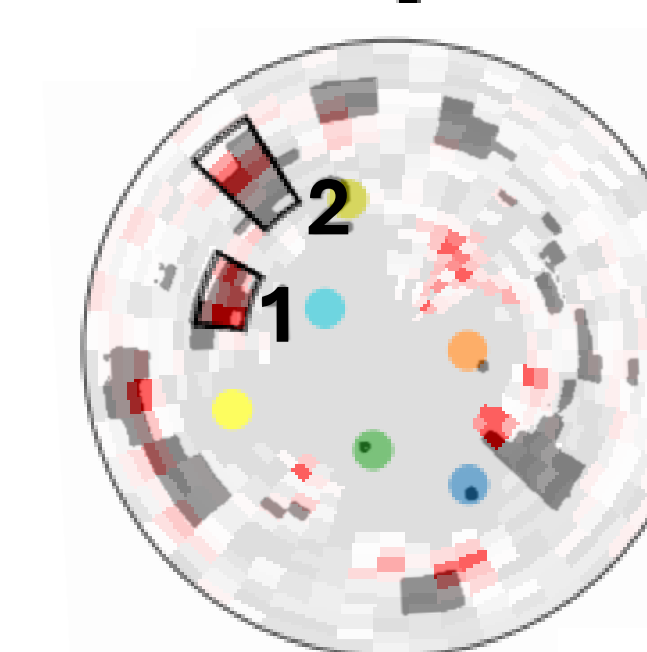


Fig. 4: CHOL headgroup density enrichment in β 2AR [3] vs. $\Delta\alpha 6$ β 2AR. To show a direct comparison, the published β 2AR result was superimposed onto the $\Delta\alpha 6$ β 2AR result, made slightly transparent, where grey corresponds to depletion and red ($\Delta\alpha 6$) or black (β 2AR [3]) corresponds to enrichment, with '1' representing the DTA threshold found in 3b. The heatmap radius is 2.5 nm. A_{site} is outlined in black. Site 1 showed similar enrichment, while site 2 showed substantially reduced enrichment. Increased CHOL density was also observed near the missing $\Delta\alpha 6$ helix.

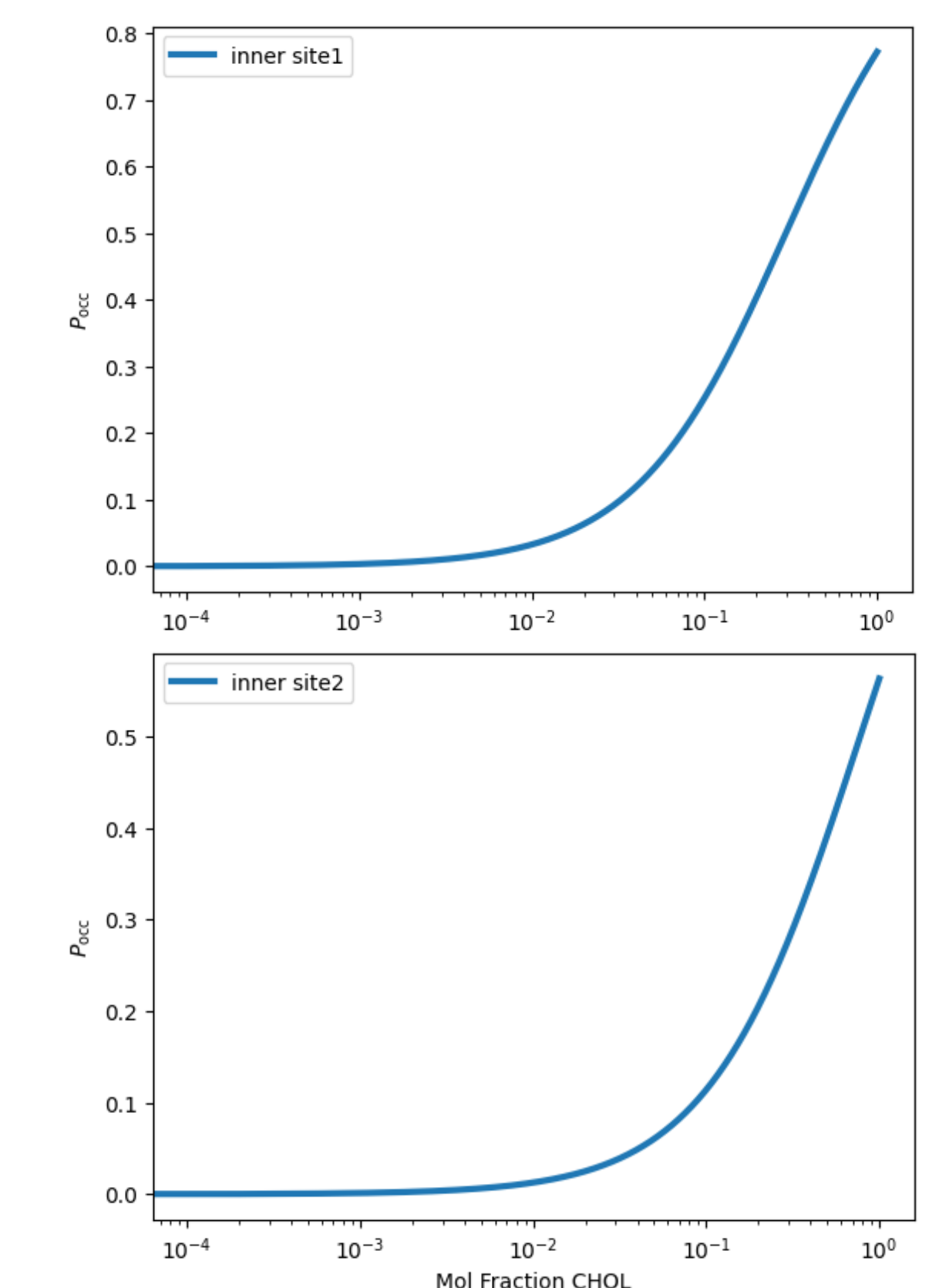


Fig. 5: CHOL occupancy probability as a function of CHOL mole fraction for sites calculated from DTA. Site 1 (top) and site 2 (bottom) show increasing occupancy probability with increasing CHOL mole fraction.

Summary

What did we do?

- Published A_{site} was used to calculate a new ΔG_{bind} using DTA for the $\Delta\alpha 6$ β 2AR variant from its trajectory.
- Compared our calculated ΔG_{bind} with published values for two CHOL-binding sites.

How well did our results match?

- Site 1: Our ΔG_{bind} -0.77 kcal/mol closely matched the published value of -0.7 ± 0.4 kcal/mol (10% error).
 - This validation supports the use of the workflow for other proteins with potential CHOL-binding sites.
- Site 2: Our ΔG_{bind} of -0.16 kcal/mol differed substantially from the published -1.4 ± 0.2 kcal/mol (89% error).
 - This difference may reflect the structural change in the $\Delta\alpha 6$ β 2AR variant, which lacks an α -helix, demonstrating that DTA can capture differences in cholesterol-binding affinity between protein conformations

Future Work: DTA applied to TCR

- β 2AR provides a validation system for applying DTA to a protein with established CHOL-binding sites.
- Successful reproduction of site 1 supports extending the approach to TCR-CHOL interactions.
- Substantial difference between site 2 supports using the approach to different TCR conformations.

References

- [1] Jaipuria, G., et al., *Cellular and Molecular Life Sciences*, 2018
- [2] Hanson, M. A., et al., *Structure*, 2008
- [3] Sandberg, J. W., et al., *Methods in Enzymology*, 2024
- [4] Chen, Y., et al., *Molecular Cell*, 2024
- [5] Chen, Y., et al., *Molecular Cell*, 2022

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Approach

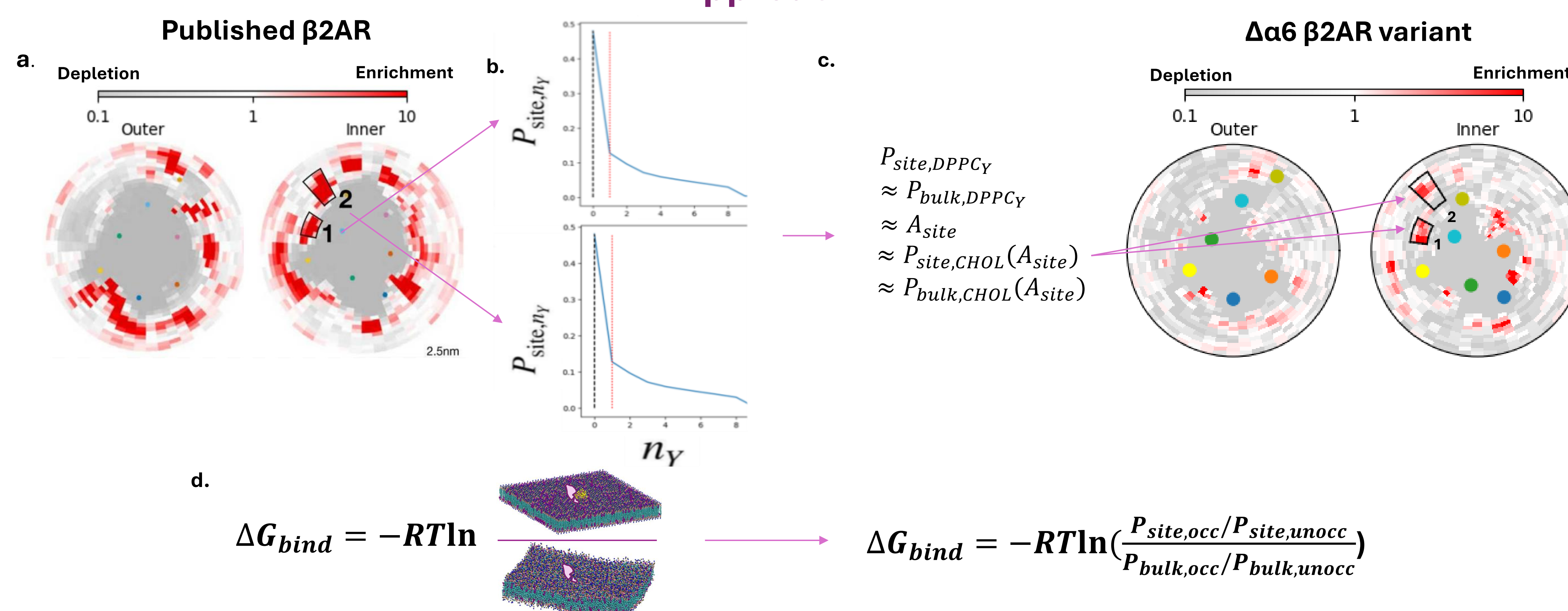


Fig. 3: Application of the published DTA workflow to the $\Delta\alpha 6$ β 2AR variant. (a) Published β 2AR CHOL density-enrichment maps, where grey corresponds to depletion and red to enrichment, identify two inner-leaflet binding sites: **site 1** and **site 2**; binding sites are outlined in black. (b) DPPC occupancy distributions (blue) define A_{site} ; $n_{peak}=0$ (black dotted) and the target peak = 1 (red dotted) define the site threshold [3]. (c) The published A_{site} was applied to the $\Delta\alpha 6$ β 2AR variant to define equivalent protein-associated and bulk-membrane regions. Note the missing helix. (d) Molecular image of 7:3 POPC:CHOL membrane with and without the β 2AR protein. Site occupancy probabilities are determined by a DTA threshold, separating occupied (occ) and unoccupied (unocc). These occupancy probabilities are used to calculate the cholesterol binding affinity (ΔG_{bind}) and CHOL occupancy across CHOL concentrations.